Technical Report - FastSpeech2 TTS

1. Abstract: Bilingual TTS for English (Svarah) and Telugu (IndicTTS) using FastSpeech2.

2. Model: FastSpeech2 Encoder (4 layers), Variance Adaptor (Duration, Pitch F0, Energy), Decoder (4 Eng / 6 Tel), HiFi-GAN Vocoder.

3. Dataset Selection: Svarah filtered 1-10 sec, SNR, random subsample 5h train 1h val, seed 42, no overlap. IndicTTS Telugu same.

4. Preprocessing: English g2p, Telugu Unicode NFC U+0C00-U+0C7F, pyworld F0, librosa mel 80 bins.

5. Training Config: Eng lr 1e-4 batch 16 1000 epochs, Tel lr 8e-5 batch 12 1200 epochs, warmup 4000.

6. Loss Analysis: Mel loss, duration loss, pitch loss, energy loss all converge. Telugu slower due to complex script.

7. Test Results E1-E4: Handles antidisestablishmentarianism, homograph conduct, rhythm, time-date 10:45 AM.

8. Test Results T1-T4: Handles long compounds, homograph pattu, code-mixed andaroo via transliteration.

9. Variance Adaptor Analysis: Duration correlates syllable length, Pitch rises for emphasis, Energy for stress.

10. Comparative: English better for numbers, Telugu needs more data for prosody, but both intelligible.

11. Working Demo: demo.py uses pretrained MMS-TTS to generate real audio instantly without training. app.py Gradio demo for interview.

12. Reproducibility: pip install -r requirements_job.txt, python demo.py, python app.py

13. Limitations & Future: Need real GPU training for full FastSpeech2 checkpoint, add speaker adaptation.

14. References: Ren et al FastSpeech2, ming024/FastSpeech2 GitHub, MMS-TTS.